

VOSSEVANGEN, Norway

WMO: 013370

Lat: 60.6250N

Lon: 6.4261E

Elev: 55

StdP: 100.67

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.3	(c) -15.9	(d) -22.8	(e) 0.5	(f) -20.4	(g) -18.4	(h) 0.7	(i) -15.5	(j) 6.1	(k) 3.0	(l) 5.4	(m) 2.4	(n) 0.4	(o) 60	(p) 0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 26.7	(d) 17.0	(e) 24.4	(f) 15.9	(g) 22.4	(h) 14.8	(i) 17.9	(j) 24.2	(k) 16.9	(l) 22.8	(m) 15.9	(n) 20.7	(o) 1.9	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.4	11.0	19.5	14.5	10.4	18.7	13.6	9.8	17.9	50.5	24.1	47.5	22.9	44.7	20.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.5	(b) 4.0	(c) 3.4	(e) -19.1	(f) 30.0	(g) 5.2	(h) 2.3	(i) -22.8	(j) 31.6	(k) -25.9	(l) 33.0	(m) -28.8	(n) 34.2	(o) -32.6	(p) 35.9		
(a) 4.5	(b) 4.0	(c) 3.4	(e) -19.4	(f) 19.2	(g) 5.2	(h) 1.8	(i) -23.1	(j) 20.5	(k) -26.1	(l) 21.5	(m) -29.0	(n) 22.5	(o) -32.7	(p) 23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-2.8	-2.3	1.3	5.6	10.1	13.6	16.6	15.1	11.4	6.3	1.8	-1.4	(6)
(7)		DBStd	7.87	6.30	5.73	4.00	2.96	3.54	2.78	3.17	2.40	2.92	3.53	4.33	5.88	(7)
(8)		HDD10.0	1931	396	343	271	136	43	3	0	0	18	123	245	352	(8)
(9)		HDD18.3	4408	654	576	529	382	257	143	75	105	208	373	495	611	(9)
(10)		CDD10.0	590	0	0	0	4	46	112	203	157	61	8	0	0	(10)
(11)		CDD18.3	26	0	0	0	0	1	2	20	3	0	0	0	0	(11)
(12)		CDH23.3	306	0	0	0	0	23	59	193	31	1	0	0	0	(12)
(13)		CDH26.7	61	0	0	0	0	4	8	46	2	0	0	0	0	(13)
(14)	Wind	WSAvg	1.5	1.6	1.5	1.6	1.6	1.5	1.4	1.4	1.3	1.5	1.5	1.5	(14)	
(15)	Precipitation	PrecAvg	2188	247	219	207	112	91	110	121	141	191	237	218	255	(15)
(16)		PrecMax	3113	490	468	607	211	203	229	199	228	342	536	439	487	(16)
(17)		PrecMin	1302	5	2	26	36	16	30	39	59	57	60	27	30	(17)
(18)		PrecStd	380	138	130	125	45	47	50	36	50	87	111	108	138	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	8.2	12.1	19.1	25.9	27.9	30.6	26.5	22.1	18.0	11.3	8.9	(19)
(20)			MCWB	4.3	4.3	6.3	10.7	15.3	16.1	18.9	17.2	14.9	12.0	8.6	5.9	(20)
(21)		2%	DB	5.7	6.5	9.0	16.3	22.9	25.0	27.7	23.8	19.5	14.7	9.8	7.0	(21)
(22)			MCWB	3.3	3.4	5.1	9.1	13.5	14.8	17.9	16.0	13.8	10.8	7.5	4.7	(22)
(23)		5%	DB	4.3	5.1	7.2	13.9	20.5	22.5	25.5	21.8	17.6	12.4	8.5	5.7	(23)
(24)			MCWB	2.3	2.5	4.1	7.7	12.2	13.6	16.9	15.2	13.0	9.7	6.7	3.7	(24)
(25)		10%	DB	3.2	3.9	6.0	11.7	17.7	20.3	23.1	19.9	16.0	11.2	7.0	4.6	(25)
(26)			MCWB	1.6	1.9	3.5	6.6	10.8	12.8	16.0	14.6	12.5	8.8	5.6	3.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	5.6	7.2	11.4	15.8	17.0	19.9	18.1	16.2	12.8	9.2	6.7	(27)
(28)			MCDWB	6.7	7.6	10.5	17.2	24.0	25.3	28.3	24.7	20.3	16.0	10.5	8.2	(28)
(29)		2%	WB	3.6	4.0	5.8	9.6	14.2	15.7	18.6	17.0	14.8	11.4	8.0	5.1	(29)
(30)			MCDWB	5.1	5.8	8.3	15.0	21.7	23.2	25.7	22.0	17.7	13.8	9.5	6.4	(30)
(31)		5%	WB	2.7	3.0	4.6	8.4	12.7	14.6	17.6	16.0	13.8	10.2	6.9	4.1	(31)
(32)			MCDWB	4.0	4.5	6.8	13.1	19.2	20.2	23.7	20.2	16.6	12.4	8.1	5.3	(32)
(33)		10%	WB	1.9	2.1	3.7	7.1	11.3	13.5	16.7	15.2	12.8	9.0	5.6	3.2	(33)
(34)			MCDWB	2.9	3.5	5.7	10.8	17.0	18.5	22.3	18.6	15.5	10.7	6.8	4.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.8	5.6	7.3	9.7	10.6	10.3	9.5	8.4	7.4	6.1	4.3	4.4	(35)
(36)			MCDBR	5.3	5.3	8.3	13.8	16.1	15.8	14.5	12.4	10.0	7.3	5.7	5.2	(36)
(37)			MCWBR	3.7	3.4	5.2	7.7	7.8	7.2	5.9	5.8	5.4	4.6	4.0	3.8	(37)
(38)		5% WB	MCDBR	4.7	5.0	7.6	12.4	14.9	12.8	13.0	10.2	8.8	6.5	5.2	4.9	(38)
(39)			MCWBR	3.6	3.6	5.2	7.2	7.4	6.3	5.5	5.2	5.2	4.5	4.1	3.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.297	0.333	0.371	0.373	0.367	0.381	0.374	0.345	0.323	0.273	0.232	(40)	
(41)		taud	2.083	2.170	2.142	2.122	2.296	2.387	2.377	2.392	2.460	2.399	2.173	2.002	(41)	
(42)		Ebn at Noon	477	680	775	814	849	862	837	813	776	661	457	335	(42)	
(43)		Edh at Noon	42	71	102	126	115	107	106	97	76	58	38	28	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.25	0.76	2.01	3.83	4.80	4.82	4.59	3.57	2.25	1.04	0.34	0.16	(44)	
(45)		RadStd	0.06	0.21	0.39	0.63	0.79	0.57	0.55	0.33	0.27	0.15	0.05	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page